MIT 18.01SC Single Variable Calculus

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1. Unit 1: Differentiation

1.1. Part A: Definition and Basic Rules

Session 1: Introduction to Derivatives

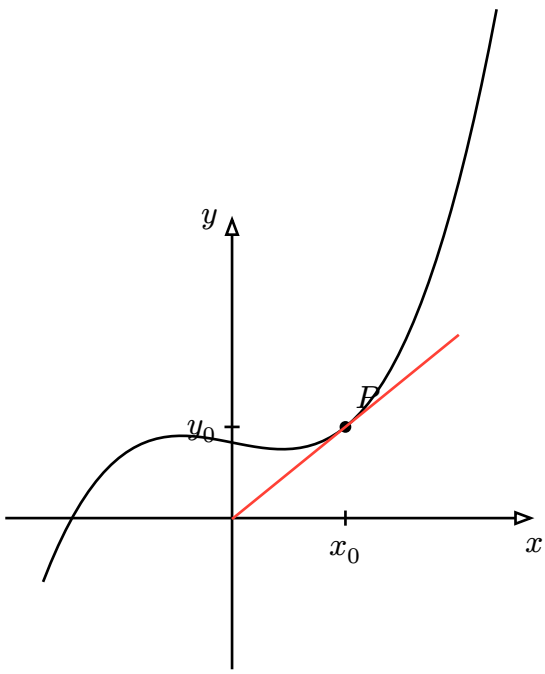
Lecture Notes

What is a dervative?

Fundamentally it's the measuring how something changes. For example, if you hold a bond and interest rates go up a little, the bond's price drops. But by how much does it drop? That sensitivity - how fast the bond's price changes as interest rates change, is a derivative.

Geometric Interpretation

Find the tangent line at $y = f(x)$ at $P = (X_o, Y_o)$ (The tangent line is the red line)



Definition 1

A tangent line is a line that approximates the direction of a curve at a specific point.

Note

Slope-intercept form: $y = mx + b$

Point-slope form: $y - y_0 = m(x - x_0)$

This is saying, if we know the point P and we know the slope m, we can write the equation of the tangent line. The point we have. The missing piece is m.

Definition 2

$f'(x_0)$, the derivative of f at x_0:

- **Geometrically:** the slope of the tangent line to $y = f(x)$ at P
- **Conceptually:** the rate of change of f at x_0

To summarize:

1. We have a curve $y = f(x)$
2. We pick a point $P = (x_0, y_0)$ on it
3. We want the tangent line at P
4. We have the formula for a line (point-slope form)
5. We know the slope $m = f'(x_0)$

Note

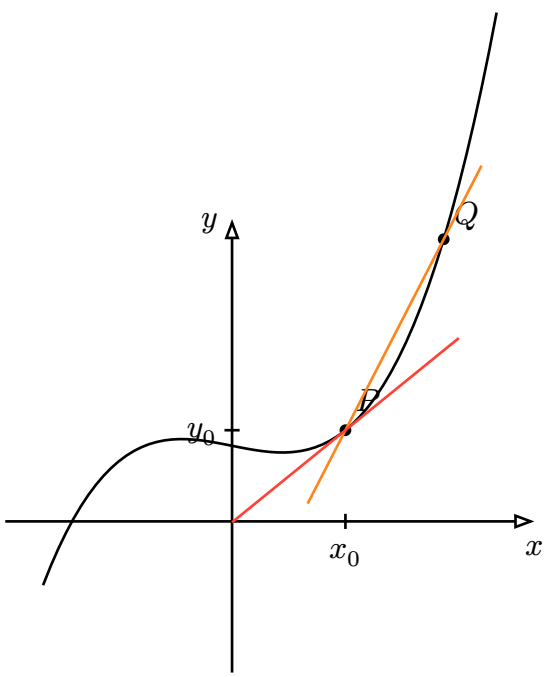
Slope is a number that measures steepness: rise over run

$$m = \frac{\Delta y}{\Delta x}$$

. A parabola is a shape (e.g. $y = x^2$). A parabola has a different slope at every point along it.

Secant Lines and Limits

(The secant line is the orange line)



Definition 3

A secant line is a straight line that connects two points on a curve. Its slope gives the average rate of change between those two points.

Note

The tangent line is the limit of secant lines. As Q approaches P, the secant line becomes the tangent line, and the average rate of change becomes the instantaneous rate of change.

Computing Slope

The slope of the secant line (average rate of change) is:

$$\frac{\Delta f}{\Delta x}$$

As $\Delta x \to 0$, the secant line becomes the tangent line:

$$m = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}$$

This limit is the derivative $f'(x_0)$

Putting it together, the definition of the derivative

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

Worked Examples

Questions

Session 2: Examples of Derivatives

Lecture Notes

Worked Examples

Questions

Session 3: Derivative as Rate of Change

Lecture Notes

Worked Examples

Questions

Session 4: Limits and Continuity

Lecture Notes

Worked Examples

Questions

Session 5: Discontinuity

Lecture Notes

Worked Examples

Questions

Session 6: Calculating Derivatives

Lecture Notes

Worked Examples

Questions

Session 7: Derivatives of Sine and Cosine

Lecture Notes

Worked Examples

Questions

Session 8: Limits of Sine and Cosine

Lecture Notes

Worked Examples

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Session 9: Product Rule

Lecture Notes

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Session 10: Quotient Rule

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Session 11: Chain Rule

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Session 12: Higher Derivatives

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1.1.1. Problem Set 1

1.2. Part B: Implicit Differentiation and Inverse Functions

Session 13: Implicit Differentiation

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Session 14: Examples of Implicit Differentiation

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Session 15: Implicit Differentiation and Inverse Functions

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Session 16: The Derivative of a^x

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Session 17: The Exponential Function, its Derivative, and its Inverse

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Session 18: Derivatives of other Exponential Functions

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Session 19: An Interesting Limit Involving e

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Session 20: Hyperbolic Trig Functions

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1.2.1. Problem Set 2

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Session 22: Materials for Exam 1

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2. Unit 2: Applications of Differentiation

2.1. Part A: Approximation and Curve Sketching

Session 23: Linear Approximation

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Session 24: Examples of Linear Approximation

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Session 25: Introduction to Quadratic Approximation

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Session 26: Using Quadratic Approximations

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Session 27: Sketching Graphs I - Polynomials and Rational Functions

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Session 28: Sketching Graphs II - General Strategies

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2.1.1. Problem Set 3

2.2. Part B: Optimization, Related Rates and Newton's Method

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Session 30: Optimization Problems II

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Session 31: Related Rates

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Session 32: Ring on a String

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Session 33: Newtons Method

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2.3. Part C: Mean Value Theorem, Antiderivatives and Differential Equations

Session 34: Introduction to the Mean Value Theorem

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Session 35: Using the Mean Value Theorem

Lecture Notes

Worked Examples

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Session 36: Differentials

Lecture Notes

Worked Examples

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Session 37: Antiderivatives

Lecture Notes

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Session 38: Integration by Substitution

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Session 39: Introduction to Differential Equations

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Session 40: Separation of Variables

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2.3.1. Problem Set 5

2.4. Exam 2

Session 41: Review for Exam 2

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Session 42: Materials for Exam 2

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3. Unit 3: The Definite Integral and its Applications

3.1. Part A: Definition of the Definite Integral and First Fundamental Theorem

Session 43: Definite Integrals

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Session 44: Adding Areas of Rectangles

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Session 45: Some Easy Integrals

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Session 46: Riemann Sums

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Session 47: Introduction of the Fundamental Theorem of Calculus

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Session 48: The Fundamental Theorem of Calculus

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Session 49: Applications of the Fundamental Theorem of Calculus

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Session 50: Combining the Fundamental Theorem and the Mean Value Theorem

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3.1.1. Problem Set 6

3.2. Part B: Second Fundamental Theorem, Areas, Volumes

Session 51: The Second Fundamental Theorem of Calculus

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Session 52: Proving the Fundamental Theorem of Calculus

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Session 53: New Functions From Old

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Session 54: The Second Fundamental Theorem and ln(x)

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Session 55: Creating New Functions Using the Second Fundamental Theorem

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Session 56: Geometric Interpretation of Definite Integrals

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Session 57: How to Calculate Volumes

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Session 58: Volume of a Sphere Revolving About x-axis

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Session 59: Volume of a Paraboloid Revolving About y-axis

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3.3. Part C: Average Value, Probability and Numerical Integration

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Session 61: Integrals and Weighted Averages

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Session 62: Integrals and Probability

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Session 63: Numerical Integration

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Session 64: Numerical Integration Continued

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Session 65: Bell Curve Conclusion

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4.1. Part A: Trigonometric Powers, Trigonometric Substitution and Completing the Square

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Session 69: Integral of sin cos Even Exponents

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Session 71: Integrals Involving secant cosecant and cotangent

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Session 72: Trig Substitution

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Session 73: Completing the Square

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4.2. Part B: Partial Fractions, Integration by Parts, Arc Length, and Surface Area

Session 74: Integration by Partial Fractions

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Session 75: Advanced Partial Fractions

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Session 76: Integration by Parts

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Session 77: Volume of a Wine Glass

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Session 78: Computing the Length of a Curve

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Session 79: Surface Area

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Session 81: Examples Using Parametrized Curves

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Session 82: Polar Coordinates

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Session 89: LHospitals Rule and Rates of Growth

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Session 91: Improper Integrals

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Session 92: Integral Comparison

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Session 96: Stacking Blocks

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Session 100: Operations on Power Series

Lecture Notes

Worked Examples

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Session 101: Conclusion

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6. Final Exam Review

6.1. Key Concepts from All Units

6.2. Practice Problems